

Shreve 《Stochastic Calculus for Finance I》 Chapter 1

二叉树无套利定价模型 / The Binomial No-Arbitrage Pricing Model

中文 | 本章用最简单的二叉树模型把**无套利定价**和**风险中性定价**两条主线讲清楚。它本身在实务里就有用（足够多期数时是连续时间模型的可计算近似），同时是后面**条件期望、鞅、连续时间模型**的入口。

EN | This chapter uses the simplest binomial model to lay out two central themes: **arbitrage pricing** and **risk-neutral pricing**. The model is useful in practice (with enough periods it is a computationally tractable approximation to continuous-time models) and is also the gateway to **conditional expectations, martingales, and continuous-time models**.

1.1 单期二叉树模型 / One-Period Binomial Model

基本设定 / Setup

- 时刻 0 股价 $S_0 > 0$ （已知）；时刻 1 取 $S_1(H)$ 或 $S_1(T)$ ，由一次抛硬币决定。*At time 0 the stock price is $S_0 > 0$ (known); at time 1 it equals $S_1(H)$ or $S_1(T)$, determined by one coin toss.*
- 上行/下行因子 / up & down factors:

$$u = \frac{S_1(H)}{S_0}, \quad d = \frac{S_1(T)}{S_0}, \quad d < u$$

直觉上 $u > 1 > d$ ，但数学上只要 $d < u$ 。*Intuitively $u > 1 > d$, but mathematically only $d < u$ is required.*

- 真实概率 $p = P(H) > 0$, $q = 1 - p > 0$, **硬币不必公平**。*Actual probabilities $p, q > 0$; the coin need **not** be fair.*
- 货币市场利率 r ：投入 \\$1 得 $1 + r$ ，借贷同利率；通常 $r \geq 0$ ，数学上只需 $r > -1$ 。
Money-market rate r : \\$1 grows to $1 + r$; same rate for borrowing and lending. Usually $r \geq 0$; mathematically only $r > -1$.

套利的定义 / Definition of Arbitrage

中文 | 套利 = 初始资金为零、亏钱概率为零、且赚钱概率为正的**交易策略**。含套利的模型不可用于分析（能凭空造钱）；真实市场偶有套利，但转瞬即逝。

EN | Arbitrage = a trading strategy that starts with zero wealth, has zero probability of losing money, and a positive probability of making money. A model admitting arbitrage is useless for

analysis (wealth from nothing). Real markets show fleeting arbitrage that disappears once exploited.

无套利条件 / No-Arbitrage Condition (1.1.2)

$$0 < d < 1 + r < u$$

为什么两个方向都必须成立 / Why both directions are needed:

- 若 $d \geq 1 + r$: 借钱买股, 即便出 T 也够还债, 且有正概率赚 $\rightarrow$ 套利。 *If $d \geq 1 + r$: borrow and buy stock; even on T the stock covers the debt, with positive probability of profit $\rightarrow$ arbitrage.*
- 若 $u \leq 1 + r$: 卖空股票投货币市场, 即便出 H 回补成本也不超过收益 $\rightarrow$ 套利。 *If $u \leq 1 + r$: short the stock and invest proceeds; even on H the replacement cost does not exceed the money-market value $\rightarrow$ arbitrage.*

反过来也成立: 满足 (1.1.2) 就一定无套利 (习题 1.1)。常取 $d = 1/u$, 但模型只需 (1.1.2)。 >

Converse holds too: (1.1.2) implies no arbitrage (Ex 1.1). Often $d = 1/u$, but only (1.1.2) is required.

用复制定价 / Pricing by Replication

中文 | 不用真实概率, 而是用股票 + 货币市场复制衍生品的收益, 复制成本就是无套利价格。

EN | Instead of using actual probabilities, **replicate** the derivative's payoff with stock + money market; the cost of replication is the no-arbitrage price.

例 / **Example 1.1.1**: $S_0 = 4$, $u = 2$, $d = \frac{1}{2}$, $r = \frac{1}{4}$, 欧式看涨 / European call $K = 5$. $S_1(H) = 8$, $S_1(T) = 2$, 期权收益 / payoff $(S_1 - K)^+ = 3$ 或/or 0。取 / take $X_0 = 1.20$, $\Delta_0 = \frac{1}{2}$ (需借 0.80 / borrow 0.80):

$$X_1(H) = \frac{1}{2}S_1(H) + (1+r)(X_0 - \Delta_0 S_0) = 3, \quad X_1(T) = 0$$

恰好匹配期权收益 $\rightarrow$ 无套利价格 = 1.20。报价偏离都可套利。 *Matches the payoff exactly $\rightarrow$ no-arbitrage price = 1.20; any other price allows arbitrage.*

一般单期公式 / General One-Period Formulas

衍生品付 $V_1(H)$ 或 $V_1(T)$ (看涨、看跌 $(K - S_1)^+$ 、远期 $S_1 - K$ 皆为特例)。 *A derivative pays $V_1(H)$ or $V_1(T)$ (call, put $(K - S_1)^+$, forward $S_1 - K$ are special cases). * 财富方程 / wealth equation:

$$X_1 = \Delta_0 S_1 + (1+r)(X_0 - \Delta_0 S_0) = (1+r)X_0 + \Delta_0(S_1 - (1+r)S_0)$$

令 $X_1(H) = V_1(H)$, $X_1(T) = V_1(T)$, 解出三件套 / solving gives three key results:

(1) 风险中性概率 / Risk-neutral probabilities (1.1.8)

$$\tilde{p} = \frac{1 + r - d}{u - d}, \quad \tilde{q} = \frac{u - 1 - r}{u - d}$$

由 (1.1.2) 知 $\tilde{p}, \tilde{q} > 0$ 且和为 1, 故可当"概率"。*By (1.1.2) both are positive and sum to 1, so they act as probabilities.*

(2) Delta 对冲 / Delta-hedging formula (1.1.9)

$$\Delta_0 = \frac{V_1(H) - V_1(T)}{S_1(H) - S_1(T)}$$

"收益之差 / 股价之差"。"Difference in payoff over difference in stock price."

(3) 风险中性定价 / Risk-neutral pricing formula (1.1.10)

$$V_0 = \frac{1}{1 + r} [\tilde{p} V_1(H) + \tilde{q} V_1(T)]$$

关键洞见 / Key Insights (务必记牢 / must remember)

- $\tilde{p}, \tilde{q}$ **不是真实概率**。真实下股票期望增长率 $>$ 货币市场 (否则没人担风险); $\tilde{p}, \tilde{q}$ 被特意选成让**股票期望增长率 = r **，从而消掉 (1.1.5) 里 Δ_0 的系数。* $\tilde{p}, \tilde{q}$ are **not** actual probabilities. Under reality the stock's mean growth exceeds the money market; $\tilde{p}, \tilde{q}$ are chosen to make the stock's expected growth equal r , eliminating the Δ_0 term.*
- **真实概率与价格无关**！对冲在涨跌两种情形都成立，与涨跌的概率无关。决定价格的是**移动幅度 u, d **。*Actual probabilities are **irrelevant** to price. The hedge works whether up or down; what matters is the **size of the moves** u, d .*
- 连续时间对应：价格依赖**波动率**，不依赖**平均增长率** (Vol II Ch.4-5)。Continuous-time analogue: prices depend on **volatility**, not on the mean rate of growth.
- 多头对冲 = 空头对冲持股数的相反数。Long-position hedge = negative of the short-position hedge.

模型假设 / Assumptions (前三条 BSM 也用 / first three also underlie BSM)

1. 股票可任意细分 / shares infinitely divisible;
2. 借贷同利率 / same borrowing & lending rate;
3. 零买卖价差 / zero bid-ask spread (实务常不成立 / often violated);
4. 每期股价只取两值 / only two values per period —— BSM 换成几何布朗运动 / replaced by geometric Brownian motion in BSM.

1.2 多期二叉树模型 / Multiperiod Binomial Model

设定与股价树 / Setup & Stock Tree

反复抛硬币，H 乘 u 、T 乘 d ；唯一假设仍是 (1.1.2)。*Toss repeatedly: H multiplies by u , T by d ; the only assumption is (1.1.2).* 例 / e.g. $S_2(HH) = u^2 S_0$, $S_2(HT) = S_2(TH) = udS_0$, $S_2(TT) = d^2 S_0$.

动态对冲 / Dynamic Hedging

每抛一次后可重新调整持仓 $\Delta_n(\omega_1 \dots \omega_n)$ (允许依赖已知抛掷结果)。*The position Δ_n can be **rebalanced after each toss**, depending on tosses observed so far.* 财富递推 / wealth recursion **(1.2.14)**:

$$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n)$$

定理 1.2.2 (多期复制定理) / Theorem 1.2.2 (Replication) —— 本章核心 / core of the chapter

给定 / given $0 < d < 1+r < u$ 与风险中性概率 $\tilde{p}, \tilde{q}$ ，对在时刻 N 付 V_N 的衍生品：

- 向后递推定价 / backward pricing recursion (1.2.16):

$$V_n(\omega_1 \dots \omega_n) = \frac{1}{1+r} [\tilde{p} V_{n+1}(\dots H) + \tilde{q} V_{n+1}(\dots T)]$$

- 逐点 Delta 对冲 / node-wise delta (1.2.17):

$$\Delta_n(\omega_1 \dots \omega_n) = \frac{V_{n+1}(\dots H) - V_{n+1}(\dots T)}{S_{n+1}(\dots H) - S_{n+1}(\dots T)}$$

- 令 $X_0 = V_0$ 并向前演化，则所有路径上 $X_N = V_N$ 。*Set $X_0 = V_0$ and roll forward $\rightarrow X_N = V_N$ on every path.*

证明思路 / proof idea：对 n 做**向前归纳** (forward induction)* 证 $X_n = V_n$ 。归纳步把 Δ_n 代入财富方程并用 V_n 定义，分别验证 H、T 两支。关键代数 / key algebra: $\frac{u - (1+r)}{u - d} = \tilde{q}$ 。

三个随机过程 / Three Stochastic Processes

(Δ_n) , (X_n) , (V_n) 均为随机过程 (按时间索引的随机变量序列)，下标 = 依赖的抛掷次数。*Each is a stochastic process (a time-indexed sequence of random variables); the subscript = number of tosses it depends on.*

完备性 / Completeness

多期二叉树是**完备市场**：任何衍生品都可复制 → 每个衍生品有**唯一**无套利价格。 *The multiperiod binomial model is **complete**: every derivative can be replicated → each has a **unique** no-arbitrage price.*

路径依赖期权也适用 / Path-Dependent Options Too

例 / **Example 1.2.4** (回望期权 / lookback) : $S_0 = 4, u = 2, d = \frac{1}{2}, r = \frac{1}{4} \Rightarrow \tilde{p} = \tilde{q} = \frac{1}{2}$, 收益 / payoff $V_3 = \max_{0 \leq n \leq 3} S_n - S_3$ 。逐节点向后递推得 $V_0 = 1.376$, $\Delta_0 = \frac{2.24 - 1.20}{8 - 2} = 0.1733$, 可对冲到时刻 3 恒等于 V_3 。

1.3 计算上的考量 / Computational Considerations

- 朴素实现 2^N 条路径 → ****指数爆炸**** (100 期约 10^{30} , 不可行)。*Naive enumeration over 2^N paths → ****exponential blow-up**** ($\approx 10^{30}$ for 100 periods).*
- 马尔可夫化简 / **Markov reduction** : 若 $V_n = v_n(S_n)$, 时刻 N 只有 $N + 1$ 个股价, 而非 2^N 条路径。

- 例 / **Ex 1.3.1** (欧式看跌 / put $K = 5$) : $v_n(s) = \frac{1}{1+r} [\tilde{p} v_{n+1}(us) + \tilde{q} v_{n+1}(ds)]$, 逐层算出 $v_0(4) = 0.864$; 持股 $\delta_n(s) = \frac{v_{n+1}(2s) - v_{n+1}(\frac{1}{2}s)}{2s - \frac{1}{2}s}$ 。

- 路径依赖降维 / **lower-dim state for path-dependence** : 用最小充分统计量。

- 例 / **Ex 1.3.2** (回望 / lookback) : 状态 (S_n, M_n) , $M_n = \max_{0 \leq k \leq n} S_k$ 。递推

$$v_n(s, m) = \frac{1}{1+r} [\tilde{p} v_{n+1}(2s, m \vee 2s) + \tilde{q} v_{n+1}(\frac{1}{2}s, m)]$$

得 $v_0(4, 4) = 1.376$ (与 Ex 1.2.4 一致 / matches Ex 1.2.4)。

要点：找到让价格成为其函数的**最小状态变量**, 把指数复杂度降为多项式。 *Key: find the **minimal state variable** of which the price is a function, turning exponential into polynomial complexity.*

1.4 小结 / Summary

- 无套利 $\iff 0 < d < 1 + r < u$, 本书始终假设之。*No arbitrage $\iff 0 < d < 1 + r < u$; always assumed.*
- 定价两种等价理解 / two equivalent views:
 - 无套利角度 / no-arbitrage** : 使"股票+货币+衍生品"市场无套利的唯一价格;
 - 对冲角度 / hedging** : 卖出后动态交易股票/货币, 使期末组合在**所有路径**上等于期权收益, 所需初始资金即价格。两者同价。

- 三件套 / the trio: 向后递推 (1.2.16)、对冲 (1.2.17)、风险中性概率 (1.2.15)。
- **风险中性概率性质 / property**：当前股价 = 下一期价格的折现风险中性平均：

$$S_n = \frac{1}{1+r} [\tilde{p} S_{n+1}(\dots H) + \tilde{q} S_{n+1}(\dots T)]$$

即在 $\tilde{p}, \tilde{q}$ 下股票期望收益率 = r = 货币市场收益率。*Under $\tilde{p}, \tilde{q}$ the stock's expected return = r = money-market rate.*

- **为什么反事实论证有效 / why the counterfactual argument is valid**：对冲必须在**所有正概率路径**上成立——解的是一组沿路径的方程，**里面没有概率**。
- 风险中性概率只是**求解这组方程的捷径**（它让“无论怎么投期望收益率都是 r ”成立）；真实概率帮不上忙，因为真实概率下组合期望收益依赖具体组合，而我们还不知道该用哪个。也可完全不提概率、只把 $\tilde{p}, \tilde{q}$ 当作 (1.2.15) 定义的数字。*The hedge must hold on **all positive-probability paths**—we solve a system of equations along paths that **contains no probabilities**. Risk-neutral probabilities are merely a **shortcut** to solving it; actual probabilities don't help because the portfolio's expected return depends on the (unknown) portfolio. One can avoid probability talk entirely and treat $\tilde{p}, \tilde{q}$ as numbers defined by (1.2.15).*

1.5 文献注记 / Notes

无套利定价隐含于 Black-Scholes, Merton 首次显式发展（以无套利为公理）；连续时间由 Harrison-Kreps、Harrison-Pliska 完善并引入**鞅**与风险中性定价。二叉树模型源自 **Cox-Ross-Rubinstein (CRR)**；期数 $\rightarrow \infty$ 时其极限重新导出 Black-Scholes-Merton（股价对数正态）。*No-arbitrage pricing is implicit in Black-Scholes, first made explicit by Merton; developed in continuous time by Harrison-Kreps and Harrison-Pliska, who introduced **martingales** and risk-neutral pricing. The binomial model is due to **Cox-Ross-Rubinstein**; its limit as periods $\rightarrow \infty$ rederives Black-Scholes-Merton (log-normal stock price).*

复习速记 / Cheat Sheet

概念 / Concept	公式 / Formula
无套利条件 / No-arbitrage	$0 < d < 1 + r < u$
风险中性概率 / Risk-neutral probs	$\tilde{p} = \frac{1 + r - d}{u - d}, \tilde{q} = \frac{u - 1 - r}{u - d}$
Delta 对冲 / Delta hedge	$\Delta_n = \frac{V_{n+1}(H) - V_{n+1}(T)}{S_{n+1}(H) - S_{n+1}(T)}$
向后递推定价 / Backward pricing	$V_n = \frac{1}{1+r} [\tilde{p} V_{n+1}(H) + \tilde{q} V_{n+1}(T)]$
财富方程 / Wealth equation	$X_{n+1} = \Delta_n S_{n+1} + (1+r)(X_n - \Delta_n S_n)$

概念 / Concept	公式 / Formula
股价鞅性质 / Martingale property	$S_n = \frac{1}{1+r} [\tilde{p} S_{n+1}(H) + \tilde{q} S_{n+1}(T)]$

****一句话本章精髓 / Chapter in one line****：衍生品价格由****复制****唯一确定；定价只依赖移动幅度 u, d （连续时间里是波动率），****与真实概率/平均增长率无关****；风险中性概率是把"沿所有路径求解对冲方程组"变简单的计算捷径。 ***A derivative's price is uniquely pinned down by ****replication****; it depends only on the move sizes u, d (volatility in continuous time), ****not on actual probabilities or mean growth****; risk-neutral probabilities are just a computational shortcut for solving the hedging system along all paths.***